

OS-FOURTH-PROGRAM > 4th_programm.py > ...

```
1  from threading import Thread, current_thread
2  from multiprocessing import Process, Pipe, Value
3
4  processes = [
5      {"pid": "P1", "arrival": 0, "burst": 7},
6      {"pid": "P2", "arrival": 2, "burst": 4},
7      {"pid": "P3", "arrival": 4, "burst": 1},
8      {"pid": "P4", "arrival": 5, "burst": 4},
9  ]
10
11  intervals = [
12      ("P1", 0, 2),
13      ("P2", 2, 4),
14      ("P1", 4, 6),
15      ("P3", 6, 7),
16      ("P2", 7, 9),
17      ("P4", 9, 11),
18      ("P1", 11, 13),
19      ("P4", 13, 15),
20      ("P1", 15, 16)
21  ]
22
23
24  def calculate_metrics():
25      print("\n--- SCHEDULING METRICS ---")
26      print("PID AT BT CT TAT WT RT")
27
28      tat_total = wt_total = rt_total = 0
29
30      for p in processes:
31          runs = [x for x in intervals if x[0] == p["pid"]]
32          first_start = runs[0][1]
33          completion = runs[-1][2]
34
35          tat = completion - p["arrival"]
36          wt = tat - p["burst"]
37          rt = first_start - p["arrival"]
38
39          tat_total += tat
40          wt_total += wt
41          rt_total += rt
42
43      print(
44          p["pid"],
45          p["arrival"],
46          p["burst"],
```

```

24  \\wsl.localhost\Ubuntu-24.04\home\shashi_linux\OS-FOURTH-PROGRAM
47      completion,
48      tat,
49      wt,
50      rt
51  )
52
53  n = len(processes)
54
55  print("Average TAT =", round(tat_total / n, 2))
56  print("Average WT =", round(wt_total / n, 2))
57  print("Average RT =", round(rt_total / n, 2))
58
59
60  def show_gantt_chart():
61      print("\n--- ROUND ROBIN GANTT CHART ---")
62
63      print(" | ".join(pid for pid, _, _ in intervals))
64
65      times = [intervals[0][1]] + [end for _, _, end in intervals]
66
67      print(" ".join(str(t) for t in times))
68
69
70  def thread_task(name):
71      print(name, "running | Thread ID:", current_thread().ident)
72
73
74  def thread_demo():
75      print("\n--- THREAD DEMO ---")
76
77      t1 = Thread(target=thread_task, args=("Thread 1",))
78      t2 = Thread(target=thread_task, args=("Thread 2",))
79
80      t1.start()
81      t2.start()
82
83      t1.join()
84      t2.join()
85
86      print("Both threads completed.")
87
88
89  def child_pipe(conn):
90      conn.send("Hello Parent - message from Child")
91      conn.close()

```

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```
92 def pipe_demo():
93     print("\n--- PIPE IPC DEMO ---")
94
95     parent_conn, child_conn = Pipe()
96
97     child = Process(
98         target=child_pipe,
99         args=(child_conn,)
100     )
101
102     child.start()
103
104     message = parent_conn.recv()
105
106     child.join()
107
108     print("Parent received:", message)
109
110
111 def update_shared(value):
112     value.value += 10
113
114
115 def shared_memory_demo():
116     print("\n--- SHARED MEMORY DEMO ---")
117
118     shared_value = Value("i", 5)
119
120     print("Before child process:", shared_value.value)
121
122     child = Process(
123         target=update_shared,
124         args=(shared_value,)
125     )
126
127     child.start()
128     child.join()
129
130     print("After child process :", shared_value.value)
131
132 if __name__ == "__main__":
133     calculate_metrics()
134     show_gantt_chart()
135     thread_demo()
136     pipe_demo()
137     shared_memory_demo()
```

```
PS Microsoft.PowerShell.Core\FileSystem::\\wsl.localhost\Ubuntu-24.04\home\shashi_linux> & C:/Python313/python.exe //wsl.localhost/Ubuntu-24.04/home/shashi_linux/OS-FOURTH-PROGRAM/4th_programm.py

--- SCHEDULING METRICS ---
PID AT BT CT TAT WT RT
P1 0 7 16 16 9 0
P2 2 4 9 7 3 0
P3 4 1 7 3 2 2
P4 5 4 15 10 6 4
Average TAT = 9.0
Average WT = 5.0
Average RT = 1.5

--- ROUND ROBIN GANTT CHART ---
P1 | P2 | P1 | P3 | P2 | P4 | P1 | P4 | P1
0 2 4 6 7 9 11 13 15 16

--- THREAD DEMO ---
Thread 1 running | Thread ID: 24144
Thread 2 running | Thread ID: 16712
Both threads completed.

--- PIPE IPC DEMO ---
Parent received: Hello Parent - message from Child

--- SHARED MEMORY DEMO ---
Before child process: 5
After child process : 15
PS Microsoft.PowerShell.Core\FileSystem::\\wsl.localhost\Ubuntu-24.04\home\shashi_linux>
```